Chapter 7

Discrete random variables and the binomial distribution

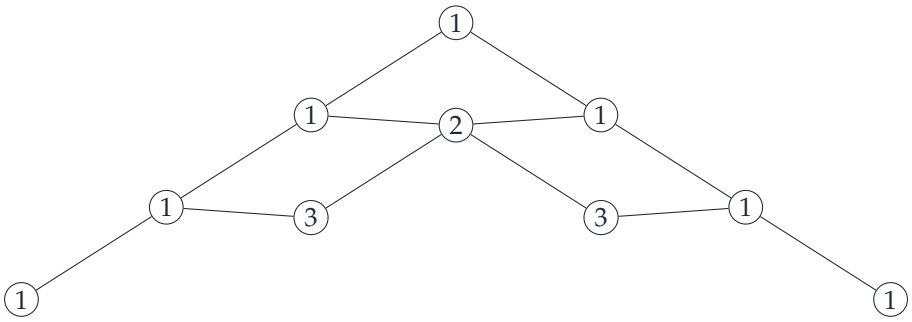
Skills: define a distribution, calculate and interpret an expected value, recognise and use a binomial distribution

7.1 Reference notes

A discrete random variable X assigns a real number to each outcome. Its distribution gives the values x_i and the probabilities $p_i = P(X = x_i)$, with $p_i \geq 0$ and $\sum p_i = 1$.

Key points

The expected value is $E(X) = \sum_i x_i p_i$. It represents the long-run average value when the experiment is repeated under the same conditions; it is not necessarily a value actually taken by X.

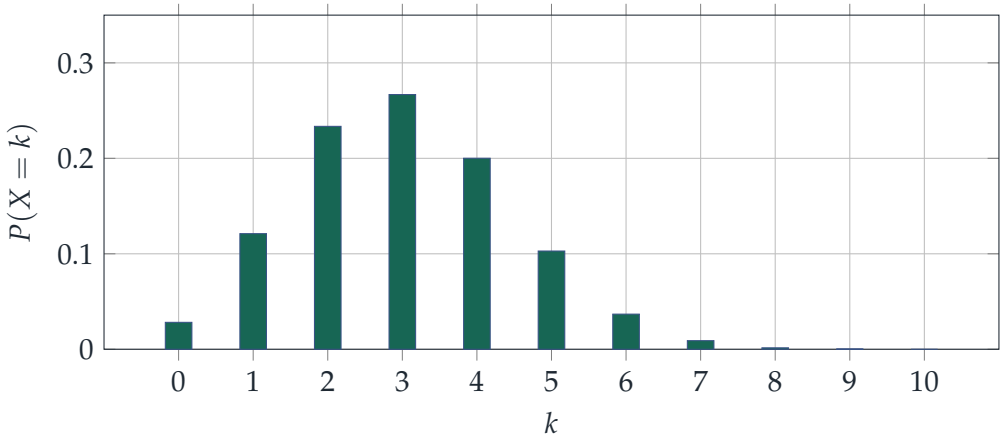


Key points

X follows a binomial distribution $\mathcal{B}(n, p)$ when X counts the number of successes in n independent Bernoulli trials with the same probability of success p. Then

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}, \qquad E(X) = np.$$

The binomial coefficient $\binom{n}{k}$ counts the number of paths containing exactly k successes among n trials. For $n \leq 10$, it can be read from Pascal's triangle.



The diagram approximately represents $\mathcal{B}(10, 0.3)$; its centre is close to $E(X) = 3$.

Session 47 — Expected value of a random variable

Discrete random variables

6 Apr 2027 • B: 2 h – C: 1 h

Session objectives

- Define a distribution.
- Calculate an expected value.
- Interpret a theoretical average.

Fluency – 6 to 8 minutes, no calculator

1. Simplify $3x + 8 - x + 2$. $2x + 10$
2. Give $E(X)$ for $X \sim \mathcal{B}(10, 0.3)$. 3
3. Calculate $\binom{5}{2}$. 10
4. Calculate the expected value of the constant X equal to 8. 8
5. Express $X \geq 1$ using a complement. $1 - P(X = 0)$

Reminder and prerequisites

Reread the last written record and be able to carry out the basic calculations needed today. Before the investigation, have students rephrase the key words: define a distribution and calculate an expected value.

Investigation

A game costs 2 euros. The gross winnings are 0, 3 or 10 euros with probabilities 0.6, 0.3, 0.1. Construct the distribution of the net gain X, calculate $E(X)$ and decide whether the game is favourable to the player.

Detailed solution – teacher edition

X takes the values $-2, 1, 8$. $E(X) = -1.2 + 0.3 + 0.8 = -0.1$ euro: the game is unfavourable on average. Have students say aloud which piece of data triggers the choice of method.

Written record

The distribution of X assigns to each value x_i the probability p_i. The expected value $E(X) = \sum x_i p_i$ is a theoretical long-run average.

Worked example

Take the data from the investigation and write the solution in order: definition of the objects, property used, exact or approximate calculation, then a concluding sentence. The model answer is in the teacher edition.

Detailed solution – teacher edition

X takes the values $-2, 1, 8$. $E(X) = -1.2 + 0.3 + 0.8 = -0.1$ euro: the game is unfavourable on average.

Practice – essential

1. Justify that the number of successes among 8 independent trials, with $p = 0.3$, follows a binomial distribution.
2. Calculate $P(X = 2)$ and $P(X = 0)$.
3. Deduce $P(X \geq 1)$ and calculate $E(X)$.

Detailed solution – teacher edition

$X \sim \mathcal{B}(8, 0.3)$. $P(X = 2) = \binom{8}{2} 0.3^2 0.7^6$, $P(X = 0) = 0.7^8$, $P(X \geq 1) = 1 - P(X = 0)$ and $E(X) = \dfrac{24}{10}$.

Textbook references

Further reading: Calao, course pp. 154–159; exercises p. 160, no. 1–5. The exercise reproduced in this session is identified below by its page and number; the reference also lets you find the figures and digital resources of the textbook.

Calao textbook – p. 160, exercise 2

A game consists of rolling a fair die. If the result is:

- 1, 2 or 3, then you lose 2 points;
- 3 or 4, then you neither lose nor win anything;
- 6, then you win 3 points.

Let G be the random variable that associates with each roll the player's gain (positive or negative).

1. Complete the table representing the probability distribution of the random variable G.
2. Calculate $E(G)$. What do you notice?

Detailed solution – teacher edition

The printed text has the result 3 appearing in two lines; the consistent partition is $1, 2, 3$ for -2, $4, 5$ for 0, and 6 for 3. We therefore obtain probabilities $1/2$, $1/3$ and $1/6$, then $E(G) = -2 \times 1/2 + 3 \times 1/6 = -0.5$. On average, the game is unfavourable to the player.

Homework

Redo practice exercise 2 with the following data: replace the initial value by 355 and the rate or common ratio by 7. Write it up in four lines: data, formula, calculation, conclusion.

Detailed solution – teacher edition

The solution depends on the exercise redone. Check above all the consistency of the model, the number of iterations, the rounding and the concluding sentence.

Exit ticket – 3 minutes

On a half-sheet: write the main rule of the session, work through a very short example and give a confidence level from 1 to 4. Essential question(s) of the day: 1, 2 and 3.

Method

1 h session: 7 min of fluency; 10 min of pooling and written record; 28 min of essential exercises; 5 min of correction; 3 min of exit ticket.
2 h session: add 20 min of investigation, 20 min of formalisation, a second independent exercise and 10 min of comparative correction.
Differentiation: give struggling students a four-box sheet (data, model, calculation, conclusion). Ask quick students to produce a counterexample, a justification or a second method.

Session 48 — Binomial distribution: recognising a situation

Discrete random variables

8 Apr 2027 • B: 1 h – C: 2 h

Session objectives

- Recognise a Bernoulli scheme.
- Identify n and p.
- Define X.

Fluency – 6 to 8 minutes, no calculator

1. Calculate 5 % of 180. 9
2. Calculate $\binom{5}{2}$. 10
3. Calculate the expected value of the constant X equal to 9. 9
4. Express $X \geq 1$ using a complement. $1 - P(X = 0)$
5. Give $E(X)$ for $X \sim \mathcal{B}(10, 0.3)$. 3

Reminder and prerequisites

Reread the last written record and be able to carry out the basic calculations needed today. Before the investigation, have students rephrase the key words: recognise a Bernoulli scheme and identify n and p.

Investigation

A customer accepts an offer with probability 0.25. An adviser contacts eight customers independently. Define X, justify the binomial distribution and identify n and p.

Detailed solution – teacher edition

X counts the acceptances; repetition of 8 identical and independent trials, success with probability 0.25: $X \sim \mathcal{B}(8, 0.25)$. Have students say aloud which piece of data triggers the choice of method.

Written record

$X \sim \mathcal{B}(n, p)$ if X counts the successes in n independent trials with the same probability p.

Worked example

Take the data from the investigation and write the solution in order: definition of the objects, property used, exact or approximate calculation, then a concluding sentence. The model answer is in the teacher edition.

Detailed solution – teacher edition

X counts the acceptances; repetition of 8 identical and independent trials, success with probability 0.25: $X \sim \mathcal{B}(8, 0.25)$.

Practice – essential

1. Justify that the number of successes among 9 independent trials, with $p = 0.3$, follows a binomial distribution.
2. Calculate $P(X = 2)$ and $P(X = 0)$.
3. Deduce $P(X \geq 1)$ and calculate $E(X)$.

Detailed solution – teacher edition

$X \sim \mathcal{B}(9, 0.3)$. $P(X = 2) = \binom{9}{2} 0.3^2 0.7^7$, $P(X = 0) = 0.7^9$, $P(X \geq 1) = 1 - P(X = 0)$ and $E(X) = \dfrac{27}{10}$.

Textbook references

Further reading: Calao, course pp. 154–159; exercises p. 160, no. 5, then p. 162, no. 14–20. The exercise reproduced in this session is identified below by its page and number; the reference also lets you find the figures and digital resources of the textbook.

Calao textbook – p. 163, exercise 22

A city's mayor has decided to hold a referendum on the possible development of a park. The results are: 63 % for and 37 % against. Two ballot papers are picked at random. The number of participants is large enough for this sampling to be treated as drawing two ballot papers at random with replacement.
Consider the random variable X that associates with each sample of two ballot papers the number of "For" ballots.

1. What distribution does X follow? Determine its parameters.
2. Calculate the probability that, in such a sample, there is: no "For"; exactly one "For".
3. Deduce from question 2 the probability that, in such a sample, there is at most one "For".

Detailed solution – teacher edition

$X \sim \mathcal{B}(2; 0.63)$. We have $P(X = 0) = 0.37^2 = 0.1369$ and $P(X = 1) = 2 \times 0.63 \times 0.37 = 0.4662$. Hence $P(X \leqslant 1) = 0.6031$.

Homework

Redo practice exercise 3 with the following data: replace the initial value by 360 and the rate or common ratio by 2. Write it up in four lines: data, formula, calculation, conclusion.

Detailed solution – teacher edition

The solution depends on the exercise redone. Check above all the consistency of the model, the number of iterations, the rounding and the concluding sentence.

Exit ticket – 3 minutes

On a half-sheet: write the main rule of the session, work through a very short example and give a confidence level from 1 to 4. Essential question(s) of the day: 1 and 2.

Method

1 h session: 7 min of fluency; 10 min of pooling and written record; 28 min of essential exercises; 5 min of correction; 3 min of exit ticket.
2 h session: add 20 min of investigation, 20 min of formalisation, a second independent exercise and 10 min of comparative correction.
Differentiation: give struggling students a four-box sheet (data, model, calculation, conclusion). Ask quick students to produce a counterexample, a justification or a second method.

Session 49 — Pascal's triangle and binomial coefficients

Discrete random variables

27 Apr 2027 • B: 2 h – C: 1 h

Session objectives

- Build Pascal's triangle.
- Read $\binom{n}{k}$.
- Interpret a coefficient.

Fluency – 6 to 8 minutes, no calculator

1. Write the multiplier for an increase of 10 %. $1 + \dfrac{10}{100} = 1.10$
2. Calculate the expected value of the constant X equal to 3. 3
3. Express $X \geq 1$ using a complement. $1 - P(X = 0)$
4. Give $E(X)$ for $X \sim \mathcal{B}(10, 0.3)$. 3
5. Calculate $\binom{5}{2}$. 10

Reminder and prerequisites

Reread the last written record and be able to carry out the basic calculations needed today. Before the investigation, have students rephrase the key words: build Pascal's triangle and read $\binom{n}{k}$.

Investigation

Build rows $n = 0$ to $n = 6$ of Pascal's triangle. Explain the calculation of the number 15 in row 6, then read $\binom{6}{2}$ and $\binom{6}{4}$.

			1			
		1		1		
	1		2		1	
1		3		3		1
1		4		6		4		1

Detailed solution – teacher edition

Each interior number is the sum of the two numbers above it. $\binom{6}{2} = \binom{6}{4} = 15$. Have students say aloud which piece of data triggers the choice of method.

Written record

Pascal's triangle satisfies $\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$. The edges equal 1.

Worked example

Take the data from the investigation and write the solution in order: definition of the objects, property used, exact or approximate calculation, then a concluding sentence. The model answer is in the teacher edition.

Detailed solution – teacher edition

Each interior number is the sum of the two numbers above it. $\binom{6}{2} = \binom{6}{4} = 15$.

Practice – essential

1. Justify that the number of successes among 10 independent trials, with $p = 0.3$, follows a binomial distribution.
2. Calculate $P(X = 2)$ and $P(X = 0)$.
3. Deduce $P(X \geq 1)$ and calculate $E(X)$.

Detailed solution – teacher edition

$X \sim \mathcal{B}(10, 0.3)$. $P(X = 2) = \binom{10}{2}0.3^2 0.7^8$, $P(X = 0) = 0.7^{10}$, $P(X \geq 1) = 1 - P(X = 0)$ and $E(X) = \dfrac{30}{10}$.

Textbook references

Further reading: Calao, p. 164, no. 24–29, then p. 166, no. 36. The exercise reproduced in this session is identified below by its page and number; the reference also lets you find the figures and digital resources of the textbook.

Calao textbook – p. 161, exercise 13

A fair coin is tossed 5 times in a row. Let F be the random variable that associates with each experiment so defined the number of "Heads" obtained.

1. F follows a binomial distribution. State its parameters.
2. Build Pascal's triangle up to $n = 5$.
3. Calculate $P(F = 3)$. Interpret this result.
4. Calculate the probability that "Heads" appears at least three times.

Detailed solution – teacher edition

$F \sim \mathcal{B}(5; 1/2)$. Row $n = 5$ of the triangle is $1, 5, 10, 10, 5, 1$. So $P(F = 3) = \binom{5}{3}(1/2)^5 = 10/32 = 0.3125$ and $P(F \geqslant 3) = (10 + 5 + 1)/32 = 0.5$.

Homework

Redo practice exercise 2 with the following data: replace the initial value by 365 and the rate or common ratio by 3. Write it up in four lines: data, formula, calculation, conclusion.

Detailed solution – teacher edition

The solution depends on the exercise redone. Check above all the consistency of the model, the number of iterations, the rounding and the concluding sentence.

Exit ticket – 3 minutes

On a half-sheet: write the main rule of the session, work through a very short example and give a confidence level from 1 to 4. Essential question(s) of the day: 1, 2 and 3.

Method

1 h session: 7 min of fluency; 10 min of pooling and written record; 28 min of essential exercises; 5 min of correction; 3 min of exit ticket.
2 h session: add 20 min of investigation, 20 min of formalisation, a second independent exercise and 10 min of comparative correction.
Differentiation: give struggling students a four-box sheet (data, model, calculation, conclusion). Ask quick students to produce a counterexample, a justification or a second method.

Session 50 — Binomial probabilities

Discrete random variables

29 Apr 2027 • B: 1 h – C: 2 h

Session objectives

- Calculate $P(X = k)$.
- Translate at least/at most.
- Use the complement.

Fluency – 6 to 8 minutes, no calculator

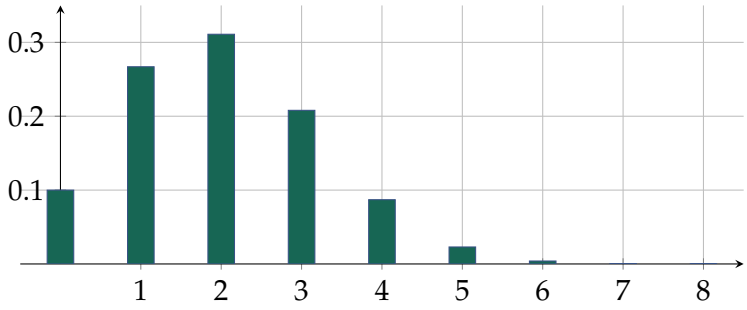
1. Simplify $3x + 4 - x + 2$. $2x + 6$
2. Express $X \geq 1$ using a complement. $1 - P(X = 0)$
3. Give $E(X)$ for $X \sim \mathcal{B}(10, 0.3)$. 3
4. Calculate $\binom{5}{2}$. 10
5. Calculate the expected value of the constant X equal to 4. 4

Reminder and prerequisites

Reread the last written record and be able to carry out the basic calculations needed today. Before the investigation, have students rephrase the key words: calculate $p(x = k)$ and translate at least/at most.

Investigation

For $X \sim \mathcal{B}(8, 0.25)$, calculate $P(X = 2)$, $P(X = 0)$, then $P(X \geq 1)$. Translate each event into a sentence.



Detailed solution – teacher edition

$P(X = 2) = \binom{8}{2}0.25^2 0.75^6 \approx 0.3115$; $P(X = 0) = 0.75^8 \approx 0.1001$; $P(X \geq 1) \approx 0.8999$. Have students say aloud which piece of data triggers the choice of method.

Written record

If $X \sim \mathcal{B}(n, p)$, then $P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}$. Translate "at least", "at most" and "none" carefully.

Worked example

Take the data from the investigation and write the solution in order: definition of the objects, property used, exact or approximate calculation, then a concluding sentence. The model answer is in the teacher edition.

Detailed solution – teacher edition

$P(X = 2) = \binom{8}{2}0.25^2 0.75^6 \approx 0.3115$; $P(X = 0) = 0.75^8 \approx 0.1001$; $P(X \geq 1) \approx 0.8999$.

Practice – essential

1. Justify that the number of successes among 6 independent trials, with $p = 0.3$, follows a binomial distribution.
2. Calculate $P(X = 2)$ and $P(X = 0)$.
3. Deduce $P(X \geq 1)$ and calculate $E(X)$.

Detailed solution – teacher edition

$X \sim \mathcal{B}(6, 0.3)$. $P(X = 2) = \binom{6}{2}0.3^2 0.7^4$, $P(X = 0) = 0.7^6$, $P(X \geq 1) = 1 - P(X = 0)$ and $E(X) = \dfrac{18}{10}$.

Textbook references

Further reading: Calao, p. 164, no. 24–29, then p. 168, no. 41–45. The exercise reproduced in this session is identified below by its page and number; the reference also lets you find the figures and digital resources of the textbook.

Calao textbook – p. 167, exercise 38

A parcel delivery company manages, for parcels collected before 10 a.m., to deliver them the next day with probability 0.75. A shopkeeper drops off seven parcels for his customers before 10 a.m. Let X be the random variable that associates with each batch of seven parcels the number of parcels delivered the next day. Assume that X follows a binomial distribution.

1. Determine the parameters n and p of the distribution of X.
2. Calculate the probability of the following events: A: “All the parcels will arrive the next day.”; B: “At least one parcel will not arrive the next day.”
3. Complete Pascal’s triangle up to row $n = 7$. Deduce $\binom{7}{2}$. Calculate the probability of the event C: “Exactly two parcels will arrive the next day.”
4. Interpret the event $\{X = 4\}$. Calculate the probability $P(X = 4)$.

Detailed solution – teacher edition

$X \sim \mathcal{B}(7; 0.75)$. $P(A) = 0.75^7 \approx 0.1335$ and $P(B) = 1 - 0.75^7 \approx 0.8665$. Since $\binom{7}{2} = 21$, $P(X = 2) = 21(0.75)^2(0.25)^5 \approx 0.0115$. $\{X = 4\}$ means that exactly four parcels are delivered the next day and $P(X = 4) = \binom{7}{4}(0.75)^4(0.25)^3 \approx 0.1730$.

Homework

Redo practice exercise 3 with the following data: replace the initial value by 370 and the rate or common ratio by 4. Write it up in four lines: data, formula, calculation, conclusion.

Detailed solution – teacher edition

The solution depends on the exercise redone. Check above all the consistency of the model, the number of iterations, the rounding and the concluding sentence.

Exit ticket – 3 minutes

On a half-sheet: write the main rule of the session, work through a very short example and give a confidence level from 1 to 4. Essential question(s) of the day: 1 and 2.

Method

1 h session: 7 min of fluency; 10 min of pooling and written record; 28 min of essential exercises; 5 min of correction; 3 min of exit ticket.
2 h session: add 20 min of investigation, 20 min of formalisation, a second independent exercise and 10 min of comparative correction.
Differentiation: give struggling students a four-box sheet (data, model, calculation, conclusion). Ask quick students to produce a counterexample, a justification or a second method.

Session 51 — Chapter review and test

Discrete random variables

4 May 2027 • B: 2 h – C: 1 h

Session objectives

- Calculate and interpret $E(X)$.
- Solve a binomial problem.
- Analyse one's mistakes.

Fluency – 6 to 8 minutes, no calculator

1. Calculate 20 % of 100. 20
2. Give $E(X)$ for $X \sim \mathcal{B}(10, 0.3)$. 3
3. Calculate $\binom{5}{2}$. 10
4. Calculate the expected value of the constant X equal to 5. 5
5. Express $X \geq 1$ using a complement. $1 - P(X = 0)$

Reminder and prerequisites

Reread the last written record and be able to carry out the basic calculations needed today. Before the investigation, have students rephrase the key words: calculate and interpret $e(x)$ and solve a binomial problem.

Investigation

A centre receives 120 groups of 10 customers, each customer complaining with probability 0.08. Estimate the average number of complaints per group, then over all the groups. Discuss what the expected value does and does not allow us to predict.

Detailed solution – teacher edition

Per group $E(X) = 10 \times 0.08 = 0.8$. Over 120 groups, about 96 complaints on average; this is not a certainty. Have students say aloud which piece of data triggers the choice of method.

Written record

For $X \sim \mathcal{B}(n, p)$, $E(X) = np$. The value gives the average number of successes over a large number of series of n trials.

Worked example

Take the data from the investigation and write the solution in order: definition of the objects, property used, exact or approximate calculation, then a concluding sentence. The model answer is in the teacher edition.

Detailed solution – teacher edition

Per group $E(X) = 10 \times 0.08 = 0.8$. Over 120 groups, about 96 complaints on average; this is not a certainty.

Practice – essential

1. Justify that the number of successes among 7 independent trials, with $p = 0.3$, follows a binomial distribution.
2. Calculate $P(X = 2)$ and $P(X = 0)$.
3. Deduce $P(X \geq 1)$ and calculate $E(X)$.

Detailed solution – teacher edition

$X \sim \mathcal{B}(7, 0.3)$. $P(X = 2) = \binom{7}{2}0.3^2 0.7^5$, $P(X = 0) = 0.7^7$, $P(X \geq 1) = 1 - P(X = 0)$ and $E(X) = \dfrac{21}{10}$.

Textbook references

Further reading: Calao, pp. 160–169. The exercise reproduced in this session is identified below by its page and number; the reference also lets you find the figures and digital resources of the textbook.

Calao textbook – p. 166, exercise 34

A company sells, in large quantities, low-energy, long-life LED bulbs. 98 % of these bulbs have an operating life of at least 20 000 h. The bulbs are sold in packs of 10. Let X be the random variable that associates with each pack of 10 bulbs the number of bulbs whose operating life is at least 20 000 h. The quantity of bulbs is large enough for the choice of a pack to be treated as a random sample with replacement of 10 bulbs.

1. What distribution does X follow? State its parameters.
2. Copy and complete: $\binom{10}{9} = \cdots$; $\binom{10}{10} = \cdots$.
3. Deduce the probabilities (rounded to 0.001): $P(X = 9)$ and $P(X = 10)$. Interpret the previous results.
4. Deduce the probability (to within 0.001) that a pack of 10 bulbs contains at least 9 bulbs whose operating life is at least 20 000 h.

Detailed solution – teacher edition

$X \sim \mathcal{B}(10; 0.98)$, $\binom{10}{9} = 10$ and $\binom{10}{10} = 1$. $P(X = 9) = 10(0.98)^9(0.02) \approx 0.167$ and $P(X = 10) = 0.98^{10} \approx 0.817$. Hence $P(X \geqslant 9) \approx 0.984$.

Homework

Redo practice exercise 2 with the following data: replace the initial value by 375 and the rate or common ratio by 5. Write it up in four lines: data, formula, calculation, conclusion.

Detailed solution – teacher edition

The solution depends on the exercise redone. Check above all the consistency of the model, the number of iterations, the rounding and the concluding sentence.

Exit ticket – 3 minutes

On a half-sheet: write the main rule of the session, work through a very short example and give a confidence level from 1 to 4. Essential question(s) of the day: 1, 2 and 3.

Method

1 h session: 7 min of fluency; 10 min of pooling and written record; 28 min of essential exercises; 5 min of correction; 3 min of exit ticket.
2 h session: add 20 min of investigation, 20 min of formalisation, a second independent exercise and 10 min of comparative correction.
Differentiation: give struggling students a four-box sheet (data, model, calculation, conclusion). Ask quick students to produce a counterexample, a justification or a second method.